Name: Mohammad Abaid

Roll no: B-08

Branch:- AI&DS(B)

Experiment:-05

AIM:- Data Visualization with Matplotlib: Visualize data using different plots like scatter, line, bar, histogram, box plot and pair plot using matplotlib and seaborn.

1.What is the importance of data visualization in data analysis? How does visualization help in EDA?

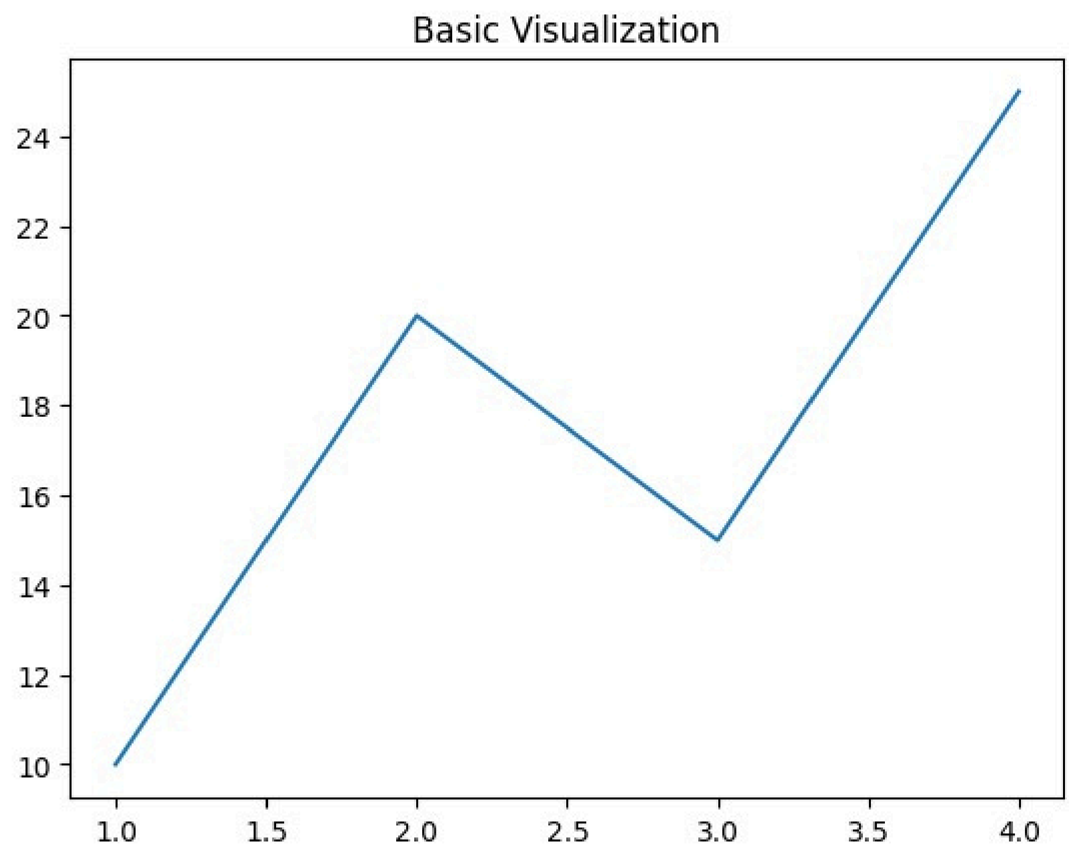
Ans:- Data visualization represents data in graphical form, which makes complex data easy to understand and interpret. It helps in quickly analyzing large datasets and improves clarity. In EDA, visualization is used to explore data and gain insights before applying any model.
It helps to:

- Identify patterns and trends
- Detect outliers
- Understand data distribution
- Compare different values
- Find relationships between variables
- Support better decision-making

```
import matplotlib.pyplot as plt

x = [1,2,3,4]
y = [10,20,15,25]

plt.plot(x,y)
plt.title("Basic Visualization")
plt.show()
```



Q2. What is the difference between Matplotlib and other visualization libraries?

ANS:- Matplotlib is a fundamental data visualization library in Python used to create static, animated, and interactive plots. It provides detailed control over every element of a graph, but requires more coding.

Other libraries like Seaborn and Plotly are built on top of Matplotlib and simplify the process of visualization by providing better default styles and advanced features.

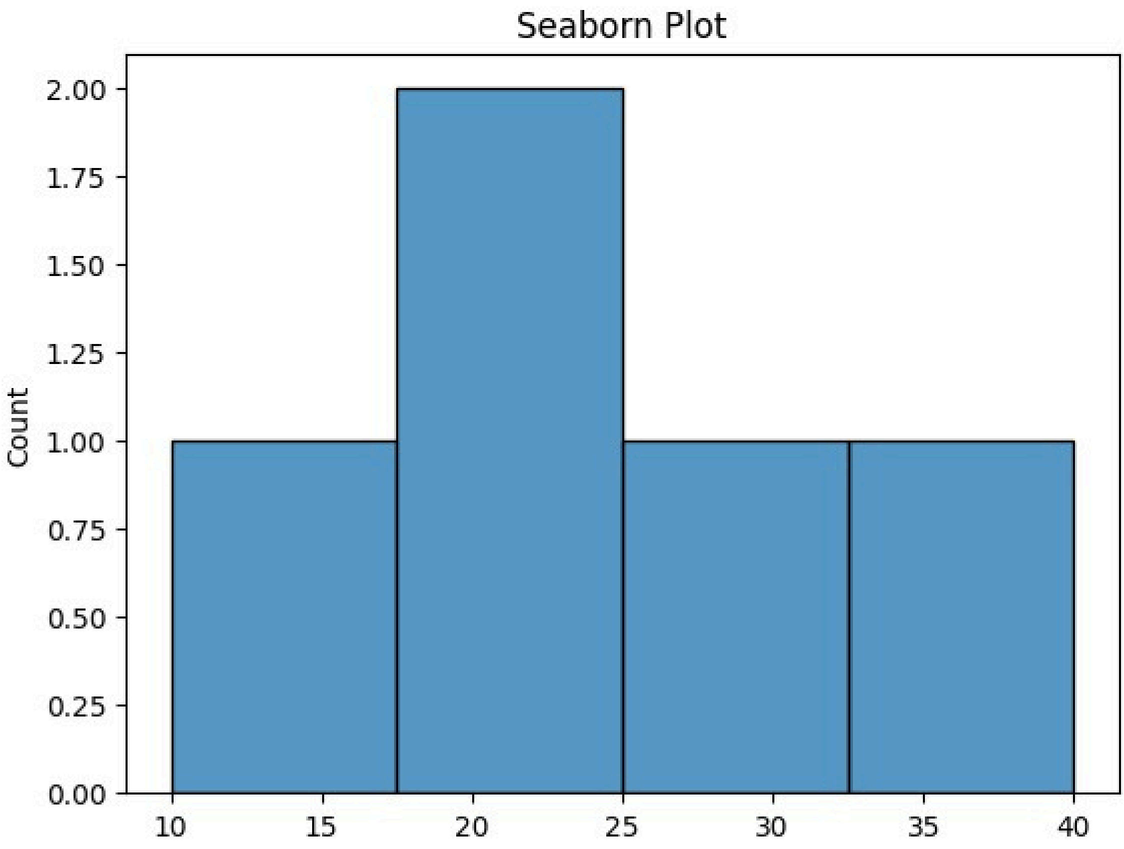
Thus, Matplotlib is mainly used when detailed customization is needed, while other libraries are preferred for quick and visually appealing plots.

- Matplotlib → Flexible and highly customizable
- Seaborn → Simple and ideal for statistical visualization
- Plotly → Interactive and web-based graphs

```
import seaborn as sns
import matplotlib.pyplot as plt

data = [10,20,20,30,40]

sns.histplot(data)
plt.title("Seaborn Plot")
plt.show()
```



Q3. What is the purpose of the pyplot module in Matplotlib?

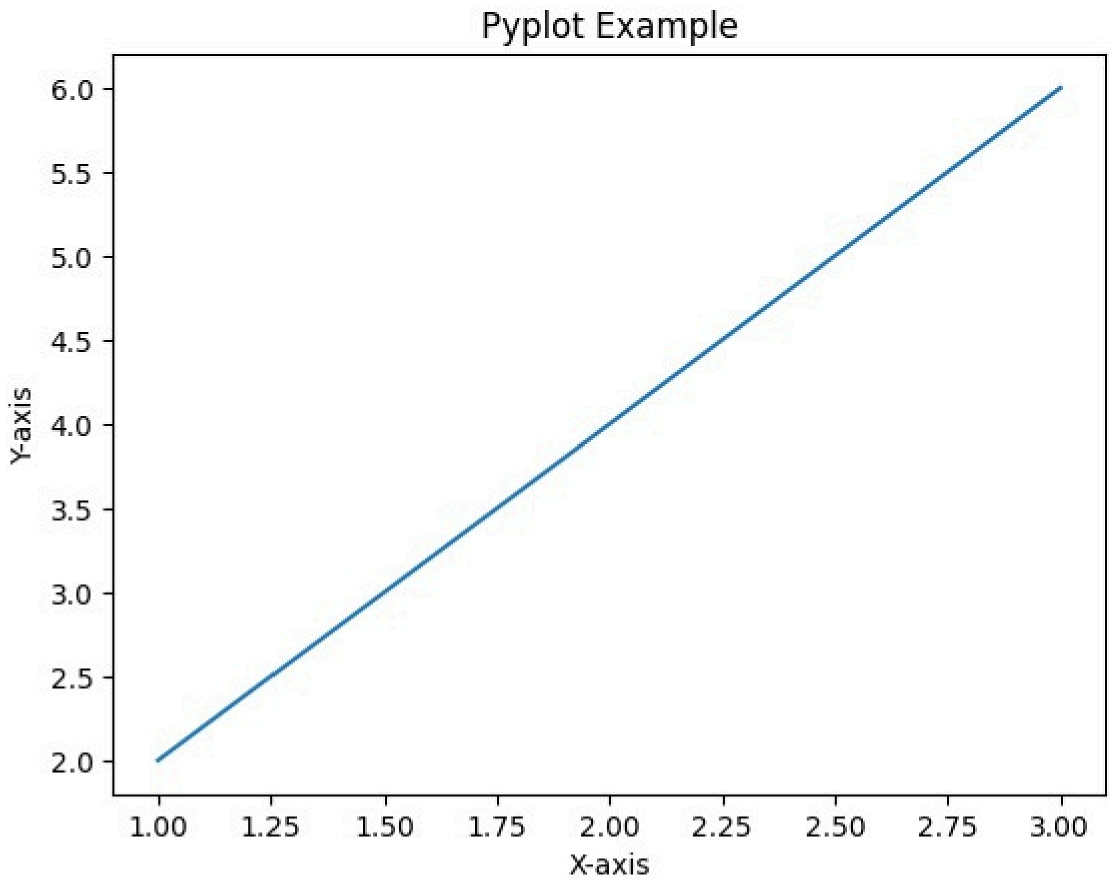
ANS:- The pyplot module is a collection of functions in Matplotlib that provides a simple interface for creating and managing plots. It allows users to create figures, plot data, and customize graphs step by step.

Pyplot is widely used because it follows a procedural approach similar to MATLAB, making it easy for beginners to understand. It also helps in controlling various components of a graph such as axes, titles, labels, legends, and grid.

Thus, pyplot acts as the main tool for visualization in Matplotlib.

```
import matplotlib.pyplot as plt

plt.plot([1,2,3],  [2,4,6])
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.title("Pyplot  Example")
plt.show()
```



Q4. Explain the process of creating a basic line plot.

ANS:- A line plot is used to represent continuous data and is very useful for showing trends and changes over time. It connects individual data points with straight lines, making it easy to observe increases or decreases.

Creating a line plot involves importing the required library, preparing the dataset, plotting the values, and enhancing the graph with labels and titles for better understanding.

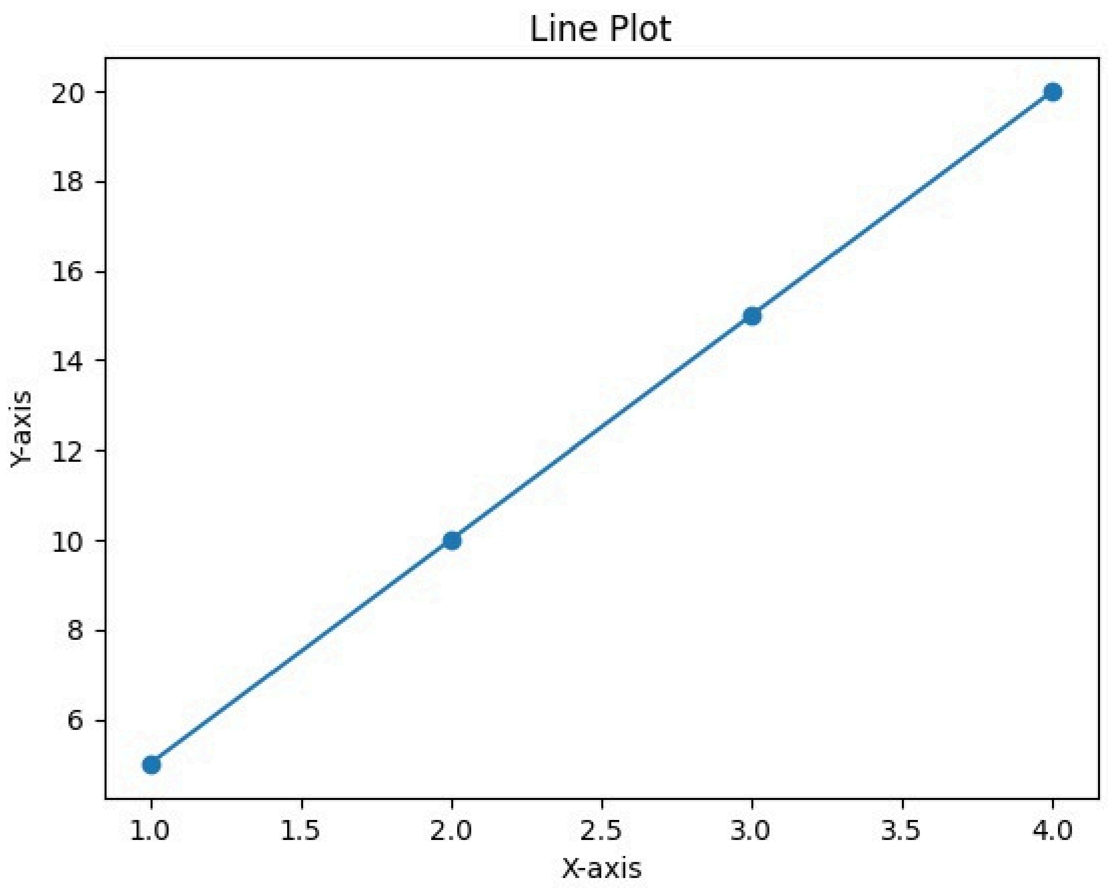
1. Import matplotlib.pyplot
2. Define x and y values
3. Use plot() function
4. Add labels and title
5. Display using show()

```
import matplotlib.pyplot as plt

x = [1,2,3,4]
y = [5,10,15,20]

plt.plot(x,y,  marker='o')
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
```

```
plt.title("Line Plot")
plt.show()
```



Q5. Why is customization important in data visualization? What customization options are available in Matplotlib?

ANS:- Customization is important in data visualization because it enhances the clarity, readability, and presentation of the data. A well-customized graph helps users to focus on important information and makes the visualization more meaningful and attractive.
Without proper customization, graphs may look confusing or difficult to interpret. Therefore, customizing elements of a plot improves communication and understanding of the data.

Matplotlib provides a wide range of customization options, allowing users to modify almost every part of a graph.

Customization options include:

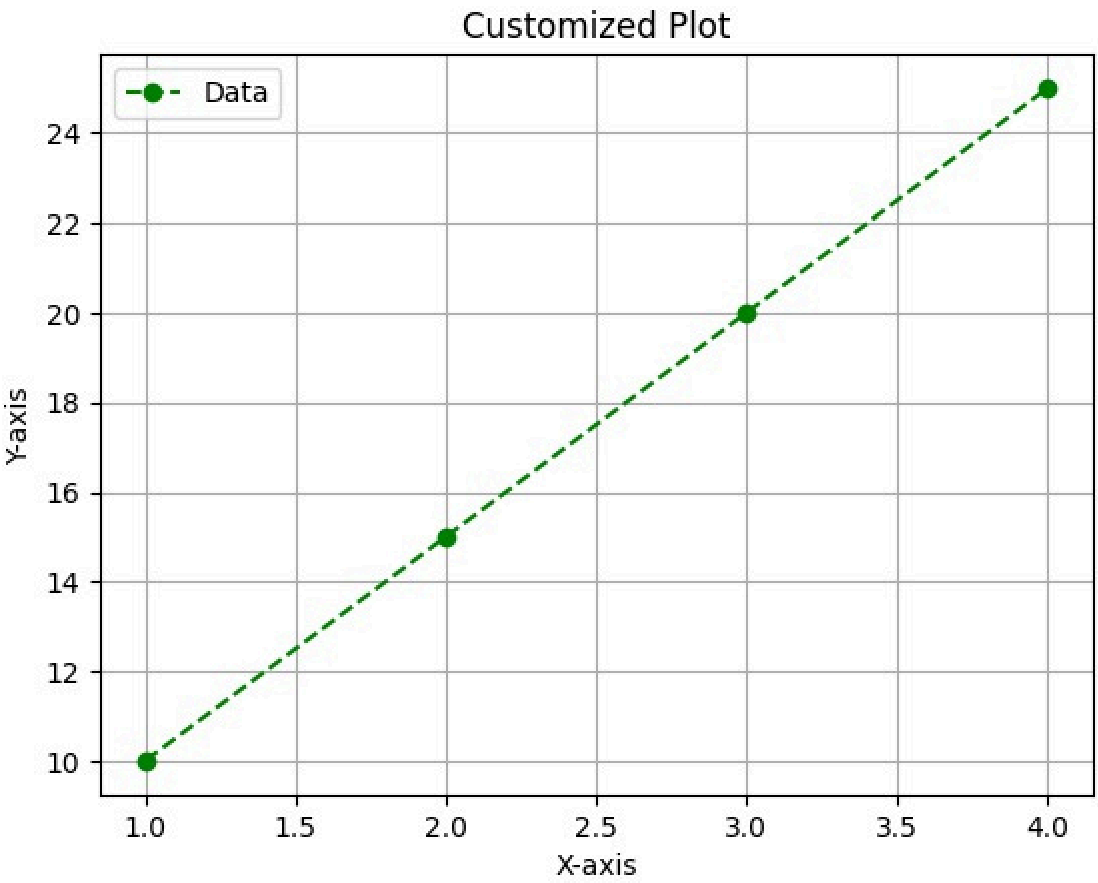
1. Changing colors, line styles, and markers
2. Adding titles and axis labels
3. Adjusting axis limits and scale
4. Modifying figure size

5. Changing font style and size
6. Adjusting axis limits and scale

```
import matplotlib.pyplot as plt

x = [1,2,3,4]
y = [10,15,20,25]

plt.plot(x,y, color='green', linestyle='--', marker='o', label='Data')
plt.title("Customized Plot") plt.xlabel("X-axis") plt.ylabel("Y-axis")
plt.legend() plt.grid() plt.show()
```



CONCLUSION:-

In this assignment, we learned the concepts and importance of data visualization in data analysis. We explored different types of plots such as line plot, scatter plot, bar plot, histogram, box plot, and pair plot using Matplotlib and Seaborn libraries. We understood how graphical representation of data helps in simplifying complex datasets and makes it easier to identify patterns, trends, relationships, and outliers. Visualization also plays a key role in Exploratory Data Analysis (EDA), as it helps in better understanding of data before applying any statistical or machine learning techniques.

Through this practical, we also learned how to customize plots by adding titles, labels, colors, legends, and grids to improve clarity and presentation. These skills are very useful in real-world applications such as data analysis, reporting, and decision-making.